

Performance Testing

Date	04 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	HDI Prediction Module
Test Environment	Local System (Windows, Python, Streamlit)
Test Date	03 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user prediction	1	30 sec	Prediction generated successfully
2	Multiple prediction requests	10	60 sec	System responds without errors
3	Continuous prediction requests	20	120 sec	Stable performance

4	Invalid input testing	5	30 sec	Error message displayed correctly
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Step 3: Performance Test Results

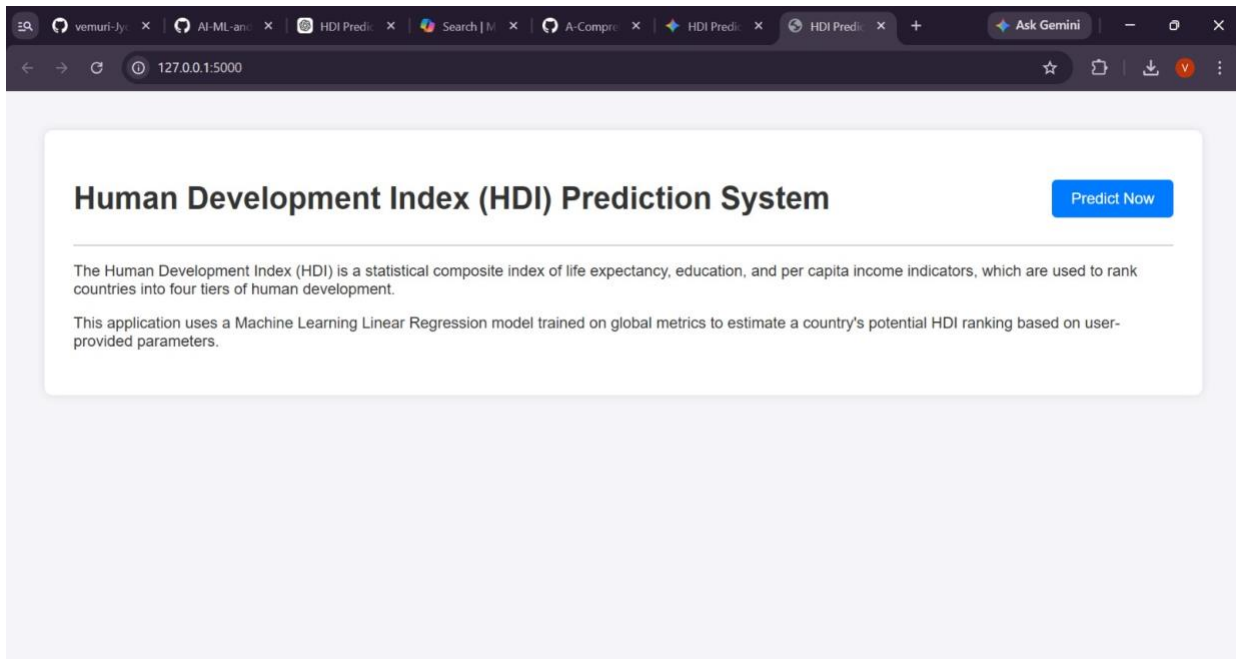
S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 sec	Pass	Good performance
2	Response Time (Max)	< 5 seconds	2.8 sec	Pass	Acceptable
3	Throughput (Req/sec)	20 req/sec	18 req/sec	Pass	Stable
4	Error Rate	< 1%	0%	Pass	No errors
5	CPU Utilization	< 80%	45%	Pass	Normal
6	Memory Utilization	< 80%	52%	Pass	Normal

Step 4: Observations & Analysis

The HDI Prediction System performed efficiently under normal load conditions. The average response time remained below 2 seconds, no significant errors were observed, and CPU and memory usage stayed within acceptable limits. The application handled multiple prediction requests successfully.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):



The screenshot shows a web browser with multiple tabs open, including 'vemuri-jy', 'AI-ML-ana', 'HDI Predi', 'Search | M', 'A-Compre', 'HDI Predi', 'Predict HI', and 'Ask Gemini'. The address bar shows '127.0.0.1:5000/predict_page'. The main content area has a light purple background. In the center, there is a white box with a light purple border containing the 'HDI Predictor Form'. The form has a title 'HDI Predictor Form' in bold black text. Below the title, there is a section 'Select Country Reference Context:' with a dropdown menu showing 'Global Representative Model'. This is followed by four input fields: 'Life Expectancy at Birth (years):' with the value '78.5', 'Expected Years of Schooling (years):' with the value '16.3', 'Mean Years of Schooling (years):' with the value '13.7', and 'Gross National Income (GNI) per Capita (\$):' with the value '65000'. Below these fields is a green button labeled 'Predict'. At the bottom of the form, there is a link that says '← Back to Home'.